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CO4 AT3 COMPARATIVE ANALYSIS

1. Compare Contiguous and Linked File Allocation

Feature	Contiguous Allocation	Linked Allocation
Storage method	File blocks are stored consecutively	File blocks can be stored anywhere and linked together
Access speed	Very fast	Slow for random access
Sequential access	Excellent	Good
Random access	Easy and fast	Difficult because previous blocks must be traversed
Memory utilization	May waste space due to external fragmentation	Better utilization; no external fragmentation
Fragmentation	External fragmentation occurs	No external fragmentation
File growth	Difficult if adjacent blocks are unavailable	Easy to add new blocks
Overhead	Low	Extra space needed for pointers

Conclusion

For sequential file access, Contiguous Allocation is more suitable because the blocks are stored next to each other, providing faster access and fewer disk movements.

However, it may suffer from external fragmentation and difficulty in file expansion.

2. Differentiate Linked and Indexed File Allocation

Feature	Linked Allocation	Indexed Allocation
Storage	Each block points to the next block	An index block contains addresses of all file blocks
Random access	Slow	Fast and direct
Sequential access	Good	Good
Storage overhead	Pointer stored in every block	Separate index block required
Reliability	Low, because losing one pointer can break the file chain	Higher reliability because block addresses are stored in the index
File size	Suitable for sequential files	Suitable for large files
Access method	Must traverse blocks one by one	Any block can be accessed directly

Conclusion

For large files, Indexed Allocation is more efficient, especially when random access is required. It allows direct access to any file block.

3. Compare FCFS and SSTF Disk Scheduling

Feature	FCFS	SSTF
Full form	First Come First Served	Shortest Seek Time First
Selection	Requests are processed in arrival order	Request closest to the current head is selected
Seek time	Usually higher	Usually lower
Performance	Lower efficiency	Higher efficiency
Fairness	Very fair	May cause starvation
Implementation	Very simple	More complex
Heavy disk load	Performance may decrease	Generally performs better

Conclusion

Under heavy disk load, SSTF generally performs better because it reduces the total head movement and average seek time.

However, FCFS is more fair because every request is served in the order of arrival, while SSTF may cause starvation for requests far from the current disk head.

4. Compare SCAN and C-SCAN Disk Scheduling

Feature	SCAN	C-SCAN
Head movement	Moves in both directions	Serves requests in only one direction
After reaching end	Reverses direction	Returns to the beginning
Service pattern	Requests on both paths are serviced	Requests serviced only during forward movement
Response time	May vary depending on position	More uniform
Fairness	Moderate	Higher
Waiting time	Can vary significantly	More predictable
Real-time suitability	Less suitable	More suitable

Conclusion

C-SCAN is generally more suitable for real-time systems because it provides a more uniform and predictable waiting time for disk requests.

SCAN may give faster service to some requests but can result in unequal waiting times.

5. Compare LOOK and C-LOOK Disk Scheduling

Feature	LOOK	C-LOOK
Direction	Serves requests in both directions	Serves requests in one direction only
End movement	Reverses at the last request	Jumps to the first pending request
Disk end	Does not move to the physical disk end	Does not move to the physical disk end
Total head movement	Usually low	Usually efficient and predictable
Response fairness	Moderate	More uniform
Implementation	Simple	Slightly more complex

Conclusion

- LOOK is better when minimizing head movement and servicing nearby requests in both directions is important.
- C-LOOK is better when a more uniform waiting time and fair service pattern are required.